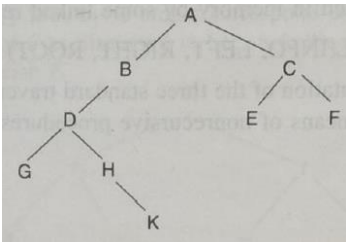


GHRCEM, Wagholi, Pune
Department of AI, SY AI A Question Bank for CAE 2
Subject – DSA

Sr. No.	Question	Unit	BL
01	Explain array representation of STACK along with algorithm for PUSH operation	3	2
02	Illustrate linked representation of STACK along with suitable example for PUSH and POP operation.	3	2
03	Solve the following arithmetic expression P written in postfix notation. P: 5, 6, 2, +, *, 12, 4, /, - (commas are used to separate the elements)	3	3
04	Find prefix notation of following infix expression 1) $(A+B)*C$ 2) $A+(B*C)$ 3) $(A+B)/(C-D)$	3	1
05	Solve the arithmetic expression P : 12, 7, 3, -, /, 2, 1, 5, +, *, +	3	3
06	Translate following infix expression Q in to its equivalent postfix expression P Q : $A + (B * C - (D / E \uparrow F) * G) * H$	3	2
07	Explain recursion with example	3	2
08	Explain Queue data structure and its Array representation	3	
09	Explain Linked representation of Queue also show insertion and deletion operation on linked representation of Queue.	3	
10	What is Deques? Write an algorithm / steps to insert an element at front and rear with suitable example.	3	1
11	Construct binary tree T of following algebraic expression E $E = (a - b) / ((c * d) + e)$	4	2
12	Define the terms related to tree by drawing simple tree structure Root Node, parent, child, siblings, descendant, ancestor, path, terminal node.	4	1
13	Explain linked representation of Binary tree in Memory with example	4	2
14	Explain sequential representation of Binary tree with example.	4	2
15	List three algorithms of traversing binary tree, explain each.	4	1
16	Demonstrate binary tree of following algebraic expression E and find out its Preorder and Postorder $E = [a + (b - c)] * [(d - e) / (f + g - h)]$	4	
17	 Find out Preorder, Inorder and Postorder of given binary tree.	4	1
18	Illustrate the six stages to insert six numbers into an empty binary tree Numbers are 40, 60, 50, 33, 55, 11	4	2
19	Explain m-Way search tree	4	2
20	Explain AVL search tree along with its representation	4	2
21	Outline tree structure and find Preorder of given Inorder Inorder : E A C K F H D B G.	4	2